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EE 446 Assignment 1 Discussion Questions

Problem 1 (e)

There is the tradeoff between the model size and accuracy. I tried some combination of distillation, pruning, and quantization. The size is smaller as I decreased each layer of the model, but the result is unstable. Therefore, based on my theory and experiment, I cannot find a proper way to further reduce size while maintaining performance.

Problem 2
(a)
Classification
Regression
Anomaly Detection (GMM)
Anomaly Detection
Anomaly Detection (K-means)

(b)
Dense: Every neuron in the layer connects to every neuron in the previous layer. Typically used near the end of a network to map extracted features to final class scores or a regression output.

1D Convolution / pooling: Detect local patterns for time-series signal where nearby points matter; down-samples the output.

2D Convolution / pooling: Similar to 1d but for data with more dimensions.

Reshape: Changes the dimension of the data without changing the actual values or their order, so that one could convert between the flat vectors (perhaps from feature extraction) and the multi-dimensional format convolutional layers expect.

Flatten: The standard bridge between convolutional layers and Dense layers since Dense layers expect flat input.

Dropout: Zeroes out a fraction of neurons on each pass, forcing the network to not rely too heavily on any single neuron. This reduces overfitting and improves generalization to new data.

(c)

Quantized (int8)

Unoptimized (float32)

(d)

Black Forest Labs Flux-Pro: Uses Replicate.com's API for the Flux Pro model to generate image.

OpenAI GPT-Image-2: Used to create synthetic images.

OpenAI Whisper: Uses the text to speech model to generate synthetic voice samples.

Synthetic data is important in machine learning applications. They fills data gaps cheaply, improves coverage of rare/edge-case scenarios, reducing blind spots, and reduces class imbalance if one category has far fewer real examples than others, improving classifier performance on underrepresented classes. It also speeds up iteration where one can generate a batch of new training examples in minutes rather than running a new physical data-collection campaign, which matters a lot when prototyping or rapidly testing whether more data actually helps.